QUIZ 2 -MTH301-page1-4

Graded

Student

YATIN PREETAM BHOJWANI

Total Points

10 / 10 pts

Question 1

Q1

5 / 5 pts

+ 0 pts Completely incorrect/ Not attempted

✔ **+ 5 pts** Completely Correct

+ 2 pts f is continuous

+ 1 pt f is bijective

+ 2 pts f^{-1} is continuous

Question 2

Q2

5 / 5 pts

+ 0 pts Completely incorrect/ Not attempted

✔ **+ 5 pts** Completely Correct

+ 2 pts Part-(a) is correct.

+ 2 pts Part-(b) is correct.

+ 1 pt Part-(c) is correct.

ANALYSIS - I : MTH 301 : QUIZ: 2025-26

Name: Yatin Preetam Bhojwani　　Roll No: 241211

- Write your name cleanly in **CAPITAL** letters and roll number inside the boxes.
- Write answers in the space provided only.　　Maximum marks: 20　　Time: 12:00 pm - 13:15 pm.

1. Let $f : (M, d) \to (N, \rho)$ be an isomorphic (i.e. $\rho(f(x), f(y)) = d(x, y)$) and onto map. Show that f is a homeomorphism.

[5]

Given: $f : (M,d) \to (N,\rho)$ & f is onto

$\rho(f(x), f(y)) = d(x,y)$

To show: f is one-one.

f is continuous f^{-1} is continuous.

let $f(x_1) = f(x_2)$

$\rho(f(x_1), f(x_2)) = d(x_1, x_2)$

$\| \; 0$

$\Rightarrow d(x_1, x_2) = 0$

$\Rightarrow x_1 = x_2$

$\Rightarrow f$ is ~~onto~~ one one

f^{-1} exists this implies $\rho(x,y) = d(f^{-1}(x), f^{-1}(y))$

take $\varepsilon > 0$

let $\delta = \varepsilon$

whenever $d(y,x) < \delta = \varepsilon$

$\rho(f(y), f(x)) = d(y,x) < \varepsilon$

$\Rightarrow f$ is continuous.

take $\varepsilon > 0$

let $\delta = \varepsilon$

whenever ~~d(y,x)~~

$\Rightarrow f^{-1}$ is continuous

$f : (M,d) \to (N,\rho)$ is bijective.

f is continuous and f^{-1} is continuous

$\Rightarrow f$ is a homeomorphism

.... Rough Work ....

f is one one

f is cont take ~~sequence~~ $d(x,y) < \varepsilon$ $\rho(f(x), f(y)) < \varepsilon$

f^{-1} is cont $\rho(f(x), f(y)) < \varepsilon$ $d(f^{-1}(x), f^{-1}(y)) < \varepsilon$.

if set is totally bounded. $\exists x_1, x_2, x_3 \ldots x_n$

M is connected

$\Rightarrow$ no A, B disjoint such that.

x_1, x_2, x_3, x_n

Can finite

Suppose M is atmost

$(x \; y \; \ldots \; x_n)$

A, B disjoint such that.

$A \cup B =$

metric space is disconnected

$\rho(y, x) < \delta = \varepsilon$

$d(f^{-1}(y), f^{-1}(x)) \leq \rho(y, x) < \varepsilon$

countable

$(x_1, x_2, \ldots$

$\Rightarrow \exists A, B$ s.t.

$A \cup B \neq M$, A

2. Let $(\ell_2, \|\cdot\|_2)$ be the complete metric space where $\ell_2 = \{(x_1, x_2, \cdots) : \sum_{n=1}^{\infty} |x_n|^2 < \infty\}$ and the norm is given by $\|x\|_2 = \left(\sum_{n=1}^{\infty} |x_n|^2\right)^{\frac{1}{2}}$ for $x = (x_1, x_2, \cdots, x_n, \cdots)$.

(a) Show that the set $A = \{x \in \ell_2 : |x_n| \leq 1/n \text{ for } n \in \mathbb{N}\}$ is a closed set in ℓ_2.

(b) Is $B = \{x \in \ell_2 : |x_n| < 1/n \text{ for } n \in \mathbb{N}\}$ an open set? Justify.

(c) Is A complete where A is defined in part (a)?

[2+2+1]

(a) $A = \{x \in \ell_2 : |x_n| \le 1$ for $n \in \mathbb{N}\}$

(b) $B = \{x \in \ell_2 : |x_n| < 1$ for $n \in \mathbb{N}\}$

hence B is not open

(c) $(\ell_2, \|\cdot\|_2)$ is complete and $A \subset \ell_2$ is closed

$\Rightarrow$ A is complete.

($A \subseteq M$ where M is complete and A is closed

$\Rightarrow$ A is complete)